bitwise avg

$$\text{ie } \frac{rax + rdx}{2} = (rax \oplus rdx) >> 1 + (rax \wedge rdx)$$

```
mov     rcx, rax
and     rcx, rdx
xor     rax, rdx
shr     rax, 1
add     rax, rcx
```

$$\left\{\begin{array}{l} rcx = rax \\ rcx = rcx \wedge rdx \\ \underline{rcx} = rax \wedge rdx \end{array}\right\}$$

$$\left\{\begin{array}{l} rax = rax \oplus rdx \\ rax_0 = rax / 2 \\ \underline{rax_0} = (rax \oplus rdx)/2 \end{array}\right\}$$

$$rax_0 = rcx + rcx$$

$$\parallel$$

$$\left\{\begin{array}{l} \underline{rcx} = rax_0 \wedge rdx \\ + \qquad + \\ \underline{rax} = (rax_0 \oplus rdx)/2 \end{array}\right\}$$

$$\left\{\begin{array}{l} \left(\frac{rax \oplus rdx}{2}\right) + (rax_0 \wedge rdx) \\ \parallel \\ (rax \oplus rdx) >> 1 + (rax \wedge rdx) \end{array}\right\}$$

```
mov     rcx, rax      ;; rax → rcx
and     rcx, rdx      ;; rcx = (rcx & rdx)
xor     rax, rdx      ;; rax := (rax ⊕ rdx)
shr     rax, 1        ;; 算术右移 logical
```

$$\text{ie } \frac{rax}{2}$$

```
add     rax, rcx
```

$$(A \gg B) + (A \mid B) \neq \frac{A + B}{2}$$

$$\leq$$

ie average of 2 values